MERKLE

Emirates

AgentOS

AI-Powered Marketing Operations Platform

Emirates Airline — Extended Overview

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Executive Summary

Why AgentOS and what it solves

Emirates operates **45+ concurrent marketing campaigns** across multiple markets, loyalty tiers, and channels. Managing this scale with traditional tools creates bottlenecks in coordination, compliance, and time-to-market. Every campaign must pass through brand validation, legal review, QA testing, and performance analysis — multiplying complexity at each step.

AgentOS is an AI-powered operations platform purpose-built for teams working with autonomous AI agents. It orchestrates a team of **15 specialized AI agents** that handle the end-to-end campaign lifecycle — from strategic planning through execution to post-launch analytics — with built-in compliance at every step.

Faster Campaign Launches

Specialized AI agents work in parallel across content, segmentation, automation, and QA — dramatically reducing the brief-to-launch cycle.

Compliance by Design

Brand, legal, and QA agents are embedded in every workflow. GDPR, UAE regulations, and premium brand tone are validated automatically.

Full Visibility & Audit Trail

Centralized dashboard with real-time agent status, daily standups, EOD reports, and a complete audit log for governance.

Intelligent Orchestration

A Campaign Intelligence Coordinator recommends agent pipelines, optimizes scheduling, and prevents conflicts across campaigns.

Skywards-Aware Targeting

CRM intelligence agent understands Emirates loyalty tiers (Blue, Silver, Gold, Platinum) for precision targeting and lifecycle automation.

Multilingual at Scale

Content agents generate premium-quality copy in English and Arabic, with personalization logic and A/B variant creation built in.

The Agent Architecture

15 specialized AI agents organized in 3 operational layers

AgentOS organizes its AI agents into three complementary layers, each with a distinct responsibility. This structure ensures that every campaign flows through strategic planning, hands-on execution, and rigorous validation before reaching the customer.

Campaign Intelligence Coordinator

The orchestration brain — routes campaigns to the right agents, recommends pipelines, generates project briefs, and maintains strategic context across all operations.

Strategic Layer — 4 Agents

Campaign planning & orchestration, CRM & Skywards intelligence, MarTech architecture, competitive monitoring. These agents define what to do and why.

Execution Layer — 5 Agents

Content creation (multilingual), audience segmentation, journey automation, send-time optimization, and email template development. These agents do the hands-on work.

Control & Validation Layer — 5 Agents

Brand compliance, legal & regulatory review, QA testing, post-campaign analytics, and documentation auditing. These agents ensure quality and compliance.

Each layer operates independently but communicates through the Coordinator, enabling parallel processing and built-in checks at every stage.

AI Agent Profiles

Detailed capabilities, tools, and workflow participation
for each of the 15 specialized agents

STRATEGIC LAYER

01 — Campaign Manager Agent

End-to-end campaign lifecycle management, KPI definition, budget optimization, and stakeholder reporting. This agent serves as the primary orchestrator for all campaign operations, coordinating work across the entire agent team and ensuring campaigns move through each phase on schedule.

KEY CAPABILITIES

Campaign Orchestration
Coordinates the full campaign lifecycle from brief creation through launch, managing dependencies between agents and ensuring smooth handoffs at each phase.

KPI Definition & Tracking
Defines measurable success criteria for each campaign — open rates, CTR, conversions, ROI — and tracks performance against targets throughout the campaign lifecycle.

Budget Optimization
Allocates campaign budgets across channels and markets, optimizing spend based on historical performance data and forecasted returns.

Cross-Channel Strategy
Plans multi-channel campaign strategies that span email, push, SMS, and in-app, ensuring consistent messaging and optimal channel selection per audience segment.

Performance Forecasting
Uses historical data and market signals to forecast campaign performance, enabling proactive adjustments before launch.

Stakeholder Reporting
Generates executive-ready campaign status reports, performance summaries, and strategic recommendations for stakeholders.

TOOLS & INTEGRATIONS

Salesforce Marketing Cloud
Campaign setup, email deployment, journey configuration, and send management

Looker Studio
Campaign performance dashboards, KPI tracking, and stakeholder reporting

Tableau
Advanced data visualization for budget allocation and ROI analysis

WORKFLOW PARTICIPATION

Campaign Creation Engine Create campaign brief, define objectives & coordinate pipeline
Flash Sale Rapid Deploy Create urgent brief with product, discount & markets
Seasonal Campaign Planning Define quarterly campaign calendar & objectives
Brand Audit Cycle Review violations & decide escalation
Email Deliverability Check Review findings & decide corrective actions
A/B Test Pipeline Decide winner & roll out
Weekly Performance Digest Add strategic commentary & recommendations

BUSINESS IMPACT

Central coordination point that prevents campaign bottlenecks, ensures deadline compliance, and provides executive visibility into the entire campaign portfolio.

STRATEGIC LAYER

02 — CRM Intelligence Agent

Loyalty and retention intelligence powered by Emirates Skywards data. This agent manages member lifecycle analysis, identity resolution across systems, and precision targeting by loyalty tier — Blue, Silver, Gold, and Platinum — to maximize member engagement and lifetime value.

KEY CAPABILITIES

CRM Segmentation
Creates sophisticated audience segments based on Skywards tier, booking history, engagement patterns, and preference data for targeted campaign delivery.

Loyalty Analytics
Analyzes Skywards program metrics — tier distribution, miles accrual patterns, redemption behavior — to identify growth and retention opportunities.

Lifecycle Automation
Designs automated member lifecycle journeys from onboarding through tier upgrades, renewal reminders, and churn prevention.

Member Scoring
Calculates engagement and propensity scores for each member, enabling predictive targeting for upgrade offers and retention campaigns.

Preference Targeting
Leverages member preferences (destinations, cabin class, meal preferences) to personalize campaign content and offers.

Data Quality Flagging
Identifies and flags data quality issues — duplicate records, stale contacts, consent gaps — to maintain CRM hygiene.

TOOLS & INTEGRATIONS

Salesforce CRM
Member profile management, contact records, and interaction history

Skywards API
Real-time member tier status, miles balance, and loyalty program data

CDP Profiles
Unified customer profiles with cross-channel identity resolution

Customer 360 Dashboard
Holistic view of member interactions across all touchpoints

WORKFLOW PARTICIPATION

Audience Hygiene Cleanup Validate contact data quality & enrich records

BUSINESS IMPACT

Enables precision targeting across 4 Skywards tiers and multiple markets, driving higher engagement rates through personalized loyalty-aware campaigns.

STRATEGIC LAYER

03 — MarTech Architecture Agent

Marketing Cloud infrastructure ownership, data model design, API integration architecture, and platform scalability. This agent ensures the technical foundation supports high-volume campaign operations without bottlenecks or failures.

KEY CAPABILITIES

Architecture Validation
Reviews and validates the technical architecture of proposed campaigns, ensuring they can be supported by current infrastructure capacity and configuration.

Data Extension Modeling
Designs and maintains the data extension structure in SFMC — subscriber tables, preference stores, transactional data — for optimal query performance.

API Integration Patterns
Architects integrations between SFMC, Skywards API, CRM, and analytics platforms using secure, scalable API patterns.

Performance Optimization
Monitors platform performance metrics (API response times, query execution, send throughput) and optimizes for peak campaign periods.

Implementation Risk Identification
Proactively identifies technical risks in campaign plans — capacity limits, data dependencies, integration failures — before they become blockers.

Technical Feasibility Assessment
Evaluates whether proposed campaign mechanics (dynamic content, real-time triggers, multi-step journeys) are technically feasible within current platform constraints.

TOOLS & INTEGRATIONS

SFMC Architecture
Platform configuration, business unit setup, and infrastructure management

Data Model Standards
Data extension schemas, naming conventions, and relationship mapping

Integration Blueprints
API integration documentation, authentication flows, and error handling patterns

Security Guidelines
Data encryption standards, access control policies, and compliance requirements

WORKFLOW PARTICIPATION

Campaign Creation Engine Validate infrastructure & capacity
Seasonal Campaign Planning Reserve infrastructure capacity for peak periods
Email Deliverability Check Validate infrastructure health & API limits
Audience Hygiene Cleanup Clean data extensions & remove stale records

BUSINESS IMPACT

Prevents infrastructure-related campaign failures, ensures platform scalability during peak periods (Ramadan, holiday season), and maintains data integrity across all integrations.

STRATEGIC LAYER

04 — Competitive Intelligence Agent

Monitors competitor airline communications across email, social media, and web channels. Identifies strategic opportunities, performs SWOT analysis, and detects emerging market trends to keep Emirates campaigns ahead of the competitive landscape.

KEY CAPABILITIES

Competitor Monitoring
Continuously tracks competitor airline marketing activities — email campaigns, social posts, website promotions — to maintain competitive awareness.

Multi-Channel Analysis
Analyzes competitor presence across email, social media, web, and paid channels to identify gaps and opportunities in Emirates' channel strategy.

SWOT Analysis
Performs structured Strengths, Weaknesses, Opportunities, and Threats analysis for each major competitor, updated on a regular cadence.

Opportunity Detection
Identifies market opportunities based on competitor weaknesses, unserved audience segments, or trending topics that Emirates can capitalize on.

Sentiment Analysis
Monitors public sentiment toward competitor brands, detecting reputation issues or positive trends that may affect Emirates' competitive positioning.

Trend Identification
Tracks industry trends in airline marketing — personalization techniques, content formats, loyalty innovations — and recommends adoption strategies.

TOOLS & INTEGRATIONS

Email Scanner
Captures and analyzes competitor email campaigns for subject lines, content, and send frequency

Social Monitor
Tracks competitor social media activity, engagement metrics, and content themes

Sentiment Analyzer
Natural language processing for brand sentiment scoring across public channels

Claude AI
Advanced analysis of competitor strategies, trend synthesis, and report generation

News Aggregator
Real-time industry news and competitor press release monitoring

WORKFLOW PARTICIPATION

Competitor Digest Weekly competitive intelligence digest with key findings and strategic recommendations

Opportunity Alerts Real-time alerts when significant competitive opportunities or threats are detected

BUSINESS IMPACT

Keeps Emirates marketing strategy informed by competitive landscape, enabling proactive rather than reactive campaign planning.

EXECUTION LAYER

05 — Content & Copywriting Agent

Creates premium, brand-compliant marketing copy in English and Arabic. Generates subject line variants, personalized email content, and A/B testing copy while maintaining Emirates' luxury brand tone across all communications.

KEY CAPABILITIES

Subject Line Generation
Generates multiple subject line variants optimized for open rates, incorporating personalization tokens, urgency cues, and brand-appropriate language.

Email Copy Drafting
Writes full email body copy including headers, body text, CTAs, and pre-headers — tailored to campaign objectives and audience segments.

Multilingual Localization (EN/AR)
Creates native-quality copy in English and Arabic, adapting tone, cultural references, and messaging for each market rather than simple translation.

Personalization Logic
Embeds dynamic personalization tokens (name, tier, destination history) and conditional content blocks based on subscriber attributes.

A/B Variant Ideation
Creates differentiated content variants for A/B testing with clear hypotheses — tone vs. urgency, short vs. long, benefit-led vs. offer-led.

Copy Optimization
Analyzes historical campaign performance to refine copy patterns, word choices, and CTA language that drive higher engagement.

TOOLS & INTEGRATIONS

Claude AI
Core LLM engine for copy generation, variant creation, and language adaptation

Emirates Prompt Library
Pre-built prompt templates for consistent brand voice across campaign types

Brand Phrases DB
Approved brand expressions, taglines, and terminology for Emirates communications

Translation Engine
Supports multilingual copy creation with cultural adaptation rules

WORKFLOW PARTICIPATION

Campaign Creation Engine Generate multilingual copy variants (EN/AR)
Re-engagement Campaign Generate "we miss you" copy & personalized offers
Flash Sale Rapid Deploy Generate multilingual copy (EN/AR) — fast track
Seasonal Campaign Planning Start creative briefs for planned campaigns
Brand Audit Cycle Correct flagged pieces & resubmit
GDPR Consent Refresh Generate re-consent request messaging
A/B Test Pipeline Generate copy variants based on hypothesis

BUSINESS IMPACT

Generates premium multilingual copy at scale, reducing content creation time while maintaining consistent brand voice across 45+ concurrent campaigns.

EXECUTION LAYER

06 — Audience Segmentation Agent

Builds audience clusters with targeting logic, suppression rules, and data extension queries. Handles tier-market combinations, audience sizing, overlap detection, and reusable segment templates for recurring campaign types.

KEY CAPABILITIES

Segment Definition
Creates precise audience segments using combinations of Skywards tier, market, booking history, engagement recency, and preference data.

Suppression Logic
Applies multi-layer suppression rules — recent purchasers, opt-outs, frequency caps, competitor exclusions — to prevent over-communication.

Audience Sizing
Calculates accurate audience counts before campaign deployment, flagging segments that are too small (low impact) or too large (low relevance).

Tier-Market Targeting
Builds combinatorial segments across 4 Skywards tiers and multiple markets (UK, UAE, US, DE) for precision targeting.

Overlap Detection
Identifies segment overlaps exceeding 30%, flagging potential audience fatigue risks and recommending deduplication strategies.

Reusable Templates
Maintains a library of reusable segment templates for recurring campaign types (route launches, holiday offers, newsletters) to accelerate future campaigns.

TOOLS & INTEGRATIONS

SFMC Data Extensions
Core data store for subscriber attributes, preferences, and transactional data

Audience Rules Library
Pre-defined segmentation rules and combinatorial logic templates

Consent & Suppression Lists
Opt-out registers, frequency caps, and regulatory suppression lists

SQL Query Builder
Custom SQL queries for complex segmentation logic against data extensions

WORKFLOW PARTICIPATION

Campaign Creation Engine Build audience segments & suppression logic
Re-engagement Campaign Build re-engagement cohort with suppression rules
Seasonal Campaign Planning Pre-build target audiences for upcoming campaigns
GDPR Consent Refresh Build affected contacts segment
Audience Hygiene Cleanup Audit segment overlaps & flag >30% duplicates

BUSINESS IMPACT

> Ensures every campaign reaches the right audience with proper suppression, preventing fatigue and maximizing engagement through precision targeting.

EXECUTION LAYER

07 — Automation Architect Agent

Designs and deploys Journey Builder flows, automation sequences, and triggered campaigns. Configures triggers, scheduling, dependency mapping, and failure handling with intelligent retry logic.

KEY CAPABILITIES

Workflow Design
Architects multi-step automation workflows with decision splits, wait steps, and conditional branching based on subscriber behavior.

Journey Scaffolding
Creates Journey Builder canvas layouts with entry sources, activities, and exits — ready for content insertion and testing.

Trigger & Scheduling
Configures event-based triggers (purchase, browse, milestone) and time-based scheduling with timezone-aware send windows.

Dependency Mapping
Maps dependencies between automation components — data feeds, API calls, content blocks — ensuring all prerequisites are met before activation.

Failure Handling & Retry
Implements intelligent retry logic for failed sends, API timeouts, and data sync errors — reducing failed sends by 34% through automated recovery.

Deployment Runbooks
Generates step-by-step deployment checklists for each journey, ensuring consistent activation and rollback procedures.

TOOLS & INTEGRATIONS

Journey Builder
Visual journey design, entry source configuration, and journey activation

Automation Studio
Scheduled automations, file imports, data extracts, and SQL activities

SFMC REST/SOAP APIs
Programmatic journey management, triggered sends, and data operations

Monitoring & Retry
Real-time journey health monitoring with automated failure recovery

WORKFLOW PARTICIPATION

Campaign Creation Engine Configure Journey Builder automation & triggers
Re-engagement Campaign Deploy drip sequence journey
GDPR Consent Refresh Deploy re-consent journey with reminders
A/B Test Pipeline Configure split test in Journey Builder

BUSINESS IMPACT

Automates complex multi-step customer journeys, reducing manual deployment effort and preventing send failures through intelligent retry mechanisms.

EXECUTION LAYER

08 — Calendar & Send-Time Agent

Optimizes send times per market and detects scheduling conflicts across the entire campaign portfolio. Enforces holiday awareness, cadence rules, and priority balancing to maximize engagement and avoid audience fatigue.

KEY CAPABILITIES

Send-Time Optimization

Analyzes historical open/click data per market to determine optimal send windows (e.g., UK: Tuesday 10am, UAE: Sunday 8pm) for maximum engagement.

Conflict Detection

Scans the campaign calendar for scheduling conflicts — overlapping sends to the same segment, competing promotions, or frequency cap violations.

Cadence Optimization

Manages communication frequency per subscriber, ensuring no contact receives more than the designated number of messages per week/month.

Holiday-Aware Scheduling

Maintains a market-specific holiday calendar (Ramadan, Christmas, National Day, Diwali) to avoid or leverage cultural moments.

Priority Balancing

When send slots conflict, applies priority rules based on campaign type, revenue potential, and business urgency to determine scheduling order.

Timeline Planning

Creates backward-scheduled timelines from launch date, calculating deadlines for each production phase (brief, content, QA, deploy).

TOOLS & INTEGRATIONS

Campaign Calendar
Central calendar showing all scheduled sends, statuses, and conflicts

Send Time Signals
Historical engagement data by market, day, and time slot

Market Holiday Calendar
Comprehensive holiday database for all Emirates markets

Peak Traffic Curves
Email platform traffic patterns to avoid high-congestion send windows

WORKFLOW PARTICIPATION

Campaign Creation Engine Schedule send date & detect conflicts

Seasonal Campaign Planning Validate dates, detect conflicts & blackout periods

BUSINESS IMPACT

Prevents send-time collisions, optimizes engagement through data-driven scheduling, and ensures cultural sensitivity across all markets.

EXECUTION LAYER

09 — Email Developer Agent

Builds responsive HTML email templates, creates reusable content blocks, and validates cross-client rendering. Maintains a modular template library for rapid campaign deployment across all device types and email clients.

KEY CAPABILITIES

Email Template Design
Creates pixel-perfect responsive email templates that render correctly across desktop, mobile, and tablet — following Emirates brand guidelines.

Responsive HTML/CSS
Writes optimized HTML and inline CSS that works across all major email clients (Gmail, Outlook, Apple Mail, Yahoo) including dark mode support.

Content Block Library
Maintains a library of reusable, modular content blocks (hero banners, product cards, CTAs, footers) for rapid template assembly.

HTML Optimization
Optimizes email HTML for file size, load time, and rendering performance — stripping unnecessary code and compressing images.

Cross-Client Rendering
Tests and validates email rendering across 90+ email client/device combinations using automated preview tools.

Template Deployment
Deploys finalized templates to SFMC Content Builder with proper naming conventions, folder structure, and version control.

TOOLS & INTEGRATIONS

SFMC Content Builder
Template storage, content block management, and deployment

HTML/CSS Validator
Code validation against email-specific HTML standards

Litmus Preview
Cross-client rendering previews across 90+ email clients and devices

Block Library
Centralized library of approved, reusable content blocks

WORKFLOW PARTICIPATION

Campaign Creation Engine Build email templates & HTML blocks
Flash Sale Rapid Deploy Assemble email from pre-approved template blocks
A/B Test Pipeline Build both template versions
Template Library Refresh Audit existing templates against current standards

BUSINESS IMPACT

Enables rapid template production with consistent rendering quality, reducing development time through reusable blocks while maintaining pixel-perfect brand presentation.

CONTROL & VALIDATION

10 — Brand Guardian Agent

Validates premium airline tone and brand compliance across all marketing communications. Enforces Emirates terminology glossary, flags casual or risky language, scores brand adherence, and ensures visual-copy alignment.

KEY CAPABILITIES

Tone Consistency
Analyzes copy for adherence to Emirates' premium tone — confident, aspirational, and sophisticated — flagging language that feels too casual, urgent, or off-brand.

Brand Compliance Scoring
Assigns a quantitative compliance score (0-100%) to each piece of content based on tone, terminology, visual alignment, and guideline adherence.

Terminology Enforcement
Validates use of approved brand terminology (e.g., "Experience" not "deal", "Premium Economy" not "premium coach") against the Emirates glossary.

Content Risk Flagging
Identifies potentially risky content — competitor mentions, price guarantees, unsubstantiated claims — before publication.

Rewrite Suggestions
Provides specific, actionable rewrite suggestions for flagged content, maintaining the original intent while aligning with brand guidelines.

Visual-Copy Alignment
Verifies that copy messaging aligns with visual creative — ensuring consistency between headlines, images, and CTAs.

TOOLS & INTEGRATIONS

Emirates Brand Guidelines
Official brand rulebook for tone, colors, typography, and imagery standards

Tone & Style Analyzer
AI-powered analysis of copy tone, readability, and brand alignment scoring

Terminology Glossary
Approved and prohibited terms database for Emirates communications

Approved Copy Library
Archive of previously approved copy for reference and pattern matching

WORKFLOW PARTICIPATION

Campaign Creation Engine Brand compliance review & tone validation
Flash Sale Rapid Deploy Fast-track brand review
Brand Audit Cycle Audit all active pieces — tone, colors, fonts, imagery
Template Library Refresh Validate templates against updated brand guidelines

BUSINESS IMPACT

Maintains a 98.5% brand compliance rate across all campaigns, protecting Emirates' premium brand positioning while enabling high-velocity content production.

CONTROL & VALIDATION

11 — Legal & Compliance Agent

Ensures regulatory compliance across GDPR, UAE data protection, CAN-SPAM, and market-specific regulations. Generates disclaimers, audits consent status, validates claims, and flags regulatory risks before campaign deployment.

KEY CAPABILITIES

Compliance Validation

Reviews campaign content against applicable regulations for each target market — GDPR (EU), UAE Federal Law, CAN-SPAM (US), CASL (Canada).

Disclaimer Generation

Automatically generates market-specific disclaimers, terms & conditions, and legal footnotes required for each campaign type and market.

Claims Scrutiny

Validates promotional claims (pricing, savings percentages, service guarantees) against verifiable data to prevent misleading advertising.

Data Privacy Checks

Verifies that personal data usage in campaigns complies with consent scope, purpose limitation, and data minimization principles.

Escalation Recommendations

When compliance issues are detected, recommends severity level and escalation path — minor (self-correct), major (legal review), critical (block send).

Audit-Ready Documentation

Generates compliance notes and decision logs for each campaign, creating an audit trail that meets regulatory inspection requirements.

TOOLS & INTEGRATIONS

Regulatory Requirements KB
Comprehensive database of market-specific regulations and requirements

Market Disclaimer Library
Pre-approved disclaimer templates for each market and campaign type

Consent & Privacy Rules
Consent scope definitions, expiration rules, and purpose limitation logic

Risk Flagging Ruleset
Automated rules for detecting high-risk content patterns and claims

WORKFLOW PARTICIPATION

Campaign Creation Engine	Legal & regulatory compliance check (GDPR, CAN-SPAM, UAE)
Flash Sale Rapid Deploy	Expedited compliance check
GDPR Consent Refresh	Detect consents approaching expiration by regulation
Audience Hygiene Cleanup	Verify consent status for all active segments

BUSINESS IMPACT

> Prevents regulatory violations that could result in significant fines (up to 4% of global turnover under GDPR), while enabling compliant campaign velocity.

CONTROL & VALIDATION

12 — QA & Deliverability Agent

Performs comprehensive pre-send validation across link checking, cross-client rendering, spam scoring, tracking validation, and content completeness. Acts as the final quality gate before any campaign goes live.

KEY CAPABILITIES

Link Validation

Checks every link in the email — UTM parameters, redirect chains, landing page status codes — ensuring zero broken links at send time.

Render Testing

Tests email rendering across major clients (Gmail, Outlook 2016/2019/365, Apple Mail, Yahoo) and devices (iOS, Android, desktop) flagging visual issues.

Content Completeness

Validates that all content slots are filled — no placeholder text, missing images, empty personalization tokens, or broken dynamic content.

Subject Line Compliance

Checks subject line length (< 50 chars optimal), preview text presence, and spam trigger word avoidance.

Spam Risk Scoring

Runs spam score analysis against major ISP filters, checking content patterns, image-to-text ratio, and authentication records (DKIM, SPF, DMARC).

QA Checklist Output

Generates a structured pass/fail QA checklist for each campaign, providing clear go/no-go status with specific failure details.

TOOLS & INTEGRATIONS

Link Checker
Automated link validation with redirect chain analysis and status code verification

Litmus / Email on Acid
Cross-client rendering previews and visual regression testing

HTML/CSS Validator
Email-specific HTML validation against coding standards

Tracking Validator
UTM parameter validation and analytics tracking verification

WORKFLOW PARTICIPATION

Campaign Creation Engine QA testing: links, renders, spam score & deliverability

Re-engagement Campaign Validate renders & deliverability

Flash Sale Rapid Deploy Rapid QA: links, renders & spam score

Email Deliverability Check Run spam score, DKIM & SPF checks across active sends

A/B Test Pipeline Validate renders for all variants

Template Library Refresh Re-test renders across email clients & deprecate broken ones

BUSINESS IMPACT

Acts as the final quality gate, catching issues that would damage deliverability, engagement, or brand perception before they reach the inbox.

CONTROL & VALIDATION

13 — Analytics & Attribution Agent

Generates post-campaign performance reports with multi-touch attribution and ROI analysis. Detects KPI anomalies, synthesizes audience insights, estimates incrementality, and produces executive summaries with next-best-action recommendations.

KEY CAPABILITIES

Post-Campaign Analysis

Produces comprehensive post-send reports covering open rates, CTR, conversion rates, revenue attribution, and benchmark comparisons within 24 hours of send.

Audience Insights Synthesis

Analyzes response patterns by segment, tier, and market to reveal which audiences over/under-performed expectations and why.

Incrementality Estimation

Estimates the true incremental impact of campaigns using holdout group analysis and statistical modeling to separate correlation from causation.

KPI Anomaly Detection

Monitors campaign metrics in real-time, automatically flagging statistically significant deviations from expected performance (±2 standard deviations).

Executive Summaries

Generates concise, stakeholder-ready summaries with key findings, performance highlights, and strategic recommendations — no data science expertise required.

Next-Best-Action Recommendations

Based on campaign results, recommends follow-up actions — retarget non-openers, expand to new segments, adjust messaging for underperformers.

TOOLS & INTEGRATIONS

GA4 / Looker Studio
Web analytics, conversion tracking, and custom performance dashboards

SFMC Engagement Logs
Email engagement data — opens, clicks, bounces, unsubscribes

CRM / Loyalty DB
Post-campaign booking and loyalty program data for revenue attribution

Tableau / PowerBI
Advanced analytics visualization and cross-channel attribution modeling

WORKFLOW PARTICIPATION

Re-engagement Campaign Detect inactive contacts (90+ days) & analyze patterns
GDPR Consent Refresh Measure opt-in rate & report compliance status
Email Deliverability Check Report bounce/complaint rate trends
A/B Test Pipeline Measure results & statistical significance
Weekly Performance Digest Compile weekly KPIs across all channels & campaigns
Audience Hygiene Cleanup Report impact on deliverability & audience metrics

BUSINESS IMPACT

> Transforms raw campaign data into actionable intelligence, enabling data-driven optimization and providing executive visibility into marketing ROI.

CONTROL & VALIDATION

14 — Documentation Auditor Agent

Audits campaign documentation for completeness, detects gaps, flags outdated documents, and scores coverage across compliance and operational documentation. Maintains audit history for governance and regulatory requirements.

KEY CAPABILITIES

Documentation Auditing

Systematically reviews all campaign-related documentation — briefs, specs, approvals, test results — against required documentation standards.

Coverage Scoring

Calculates a documentation coverage score per campaign and per department, identifying areas where documentation falls below acceptable thresholds.

Gap Detection

Identifies missing documentation — unsigned approvals, absent test reports, incomplete compliance records — and generates remediation tasks.

Outdated Document Flagging

Flags documents that haven't been updated within their review period, ensuring all active documentation reflects current processes and requirements.

Audit History

Maintains a complete audit trail of all documentation reviews, scores, and remediation actions for regulatory inspection readiness.

Compliance Documentation

Ensures all regulatory-required documentation (consent records, data processing agreements, impact assessments) is present and current.

TOOLS & INTEGRATIONS

Claude AI
Document analysis, gap detection, and coverage scoring through natural language understanding

Confluence
Documentation storage, retrieval, and version history tracking

WORKFLOW PARTICIPATION

Documentation Audit Full documentation audit cycle with scoring, gap detection, and remediation tracking

BUSINESS IMPACT

Ensures campaign governance through systematic documentation coverage, reducing audit preparation time and maintaining regulatory readiness.

ORCHESTRATION

15 — Campaign Intelligence Coordinator

The strategic brain of AgentOS. This meta-agent orchestrates all 14 operational agents, maintains context across the entire campaign portfolio, and acts as the primary interface for human operators. It refines campaign ideas through natural language conversation, recommends optimal agent pipelines, classifies campaigns into 29 BAU types, and generates detailed project specifications.

KEY CAPABILITIES

Agent Orchestration
Routes campaign requests to the appropriate agents based on campaign type, complexity, and current agent workload — ensuring optimal resource allocation.

Pipeline Recommendation
Analyzes campaign requirements and recommends the optimal agent pipeline — which agents are needed, in what order, and with what dependencies.

Brief Generation
Generates detailed campaign project specifications from natural language conversation — objectives, target audience, messaging, timeline, and success metrics.

BAU Classification
Classifies campaigns into 29 Business-As-Usual types (Route Launch, Holiday Offer, Partner Launch, Newsletter, etc.) to apply proven playbooks and templates.

Strategic Context Maintenance
Maintains awareness of all active campaigns, agent statuses, and portfolio-level metrics to make informed orchestration decisions.

Natural Language Interface
Enables human operators to interact with the entire agent system through conversational AI, lowering the barrier to campaign planning and execution.

TOOLS & INTEGRATIONS

Claude AI (Sonnet 4.6)
Core reasoning engine for conversation, analysis, and project generation

All 14 Operational Agents
Full orchestration access to delegate tasks and coordinate workflows

Campaign Portfolio Database
Historical and active campaign data for context-aware recommendations

Workspace Context
Team structure, department organization, and operational parameters

WORKFLOW PARTICIPATION

All Workflows Orchestrates, monitors, and coordinates agent pipelines across all 11 defined workflows

BUSINESS IMPACT

Enables teams to go from campaign idea to structured execution plan in minutes through natural language conversation, dramatically reducing planning overhead.

Workflow Deep Dives

Step-by-step breakdowns of all 11 automated workflows that orchestrate multi-agent collaboration

Campaign Creation Engine

End-to-end campaign production pipeline that takes a campaign from initial brief to live deployment. This is the core workflow that coordinates the most agents and represents the primary value chain of the system.

TRIGGER
New campaign request from stakeholder or generated from weekly planning session

OUTPUT
Fully deployed, QA-validated campaign with all compliance approvals, live in Journey Builder

AGENTS INVOLVED
10 agents: Campaign Manager Agent, Segmentation Agent, Content Agent, Email Developer Agent, Brand Guardian Agent, Legal Agent, Calendar Agent, Automation Architect Agent, MarTech Architecture Agent, QA Agent

STEP-BY-STEP BREAKDOWN

1. **Campaign Manager Agent**
   Create campaign brief, define objectives, KPIs, target markets, and budget allocation. Coordinate the full agent pipeline.
2. **Segmentation Agent**
   Build audience segments with Skywards tier targeting and multi-layer suppression rules. Size audiences and validate data quality.
3. **Content Agent**
   Generate multilingual copy variants (EN/AR) — subject lines, body copy, CTAs, and pre-headers with personalization tokens.
4. **Email Developer Agent**
   Build responsive email templates using content blocks. Assemble final creative with dynamic content zones.
5. **Brand Guardian Agent**
   Brand compliance review — validate tone, terminology, visual-copy alignment. Score compliance and flag violations.
6. **Legal Agent**
   Regulatory compliance check — GDPR, CAN-SPAM, UAE law. Generate market-specific disclaimers and validate claims.
7. **Calendar Agent**
   Schedule optimal send date and time per market. Detect conflicts with other campaigns and enforce cadence rules.
8. **Automation Architect Agent**
   Configure Journey Builder flow — entry sources, decision splits, wait steps, and triggered sends.
9. **MarTech Architecture Agent**
   Validate infrastructure capacity — API limits, data extension performance, send throughput for expected volume.
10. **QA Agent**
    Final quality gate — link validation, cross-client rendering (90+ clients), spam scoring, tracking verification, and go/no-go checklist.

Re-engagement Campaign

Automated win-back workflow for contacts inactive for 90+ days. Uses behavioral analysis to identify at-risk contacts and deploys targeted drip sequences to re-engage them before they churn.

TRIGGER
Analytics agent detects contacts with 90+ days of inactivity

OUTPUT
Multi-step drip sequence deployed via Journey Builder targeting inactive contacts

AGENTS INVOLVED
5 agents: Analytics Agent, Segmentation Agent, Content Agent, QA Agent, Automation Architect Agent

STEP-BY-STEP BREAKDOWN

1. **Analytics Agent**
   Detect inactive contacts (90+ days of no opens, clicks, or site visits). Analyze inactivity patterns to identify recoverable vs. lost contacts.
2. **Segmentation Agent**
   Build re-engagement cohort with layered suppression — exclude recent purchasers, active complainers, and hard bounces.
3. **Content Agent**
   Generate personalized "we miss you" copy with tiered offers based on member value (exclusive fares for Gold/Platinum, miles bonuses for Blue/Silver).
4. **QA Agent**
   Validate email renders across clients, check all links, verify personalization tokens render correctly for each tier variant.
5. **Automation Architect Agent**
   Deploy multi-step drip sequence — initial reach-out, 7-day reminder, final offer with expiration — with behavioral triggers for early conversion.

Flash Sale Rapid Deploy

Emergency campaign deployment workflow designed for time-sensitive promotions. Uses pre-approved templates and fast-track review processes to achieve launch in under 4 hours.

TRIGGER
Urgent business request for time-sensitive promotion (flash sale, partner deal, competitive response)

OUTPUT
Live campaign deployed within 4 hours using pre-approved templates and expedited reviews

AGENTS INVOLVED
6 agents: Campaign Manager Agent, Content Agent, Email Developer Agent, Brand Guardian Agent, Legal Agent, QA Agent

STEP-BY-STEP BREAKDOWN

1. **Campaign Manager Agent**
   Create urgent brief specifying product, discount percentage, target markets, and promotion window. Flag as high-priority.
2. **Content Agent**
   Fast-track multilingual copy generation (EN/AR) using pre-approved flash sale prompt templates for consistent brand voice under time pressure.
3. **Email Developer Agent**
   Rapid template assembly using pre-approved content blocks — swap hero image, update offer details, configure countdown timer.
4. **Brand Guardian Agent**
   Fast-track brand review — focused on critical items (tone, claims, imagery) with relaxed checks on minor style elements.
5. **Legal Agent**
   Expedited compliance check — validate pricing claims, generate required disclaimers, check offer terms against regulations.
6. **QA Agent**
   Rapid QA pass — critical link checks, primary client renders (Gmail, Outlook, Apple Mail), spam score verification.

Seasonal Campaign Planning

Quarterly strategic planning workflow that defines the campaign calendar, pre-builds audiences, starts creative briefs, and reserves infrastructure capacity for upcoming peak periods (Ramadan, summer, holiday season).

TRIGGER
Quarterly planning cycle (typically 6-8 weeks before season start)

OUTPUT
Complete quarterly campaign calendar with pre-built audiences, draft briefs, and reserved infrastructure

AGENTS INVOLVED
5 agents: Campaign Manager Agent, Calendar Agent, Segmentation Agent, Content Agent, MarTech Architecture Agent

STEP-BY-STEP BREAKDOWN

1. **Campaign Manager Agent**
   Define quarterly campaign calendar — map all planned campaigns against key dates, allocate budgets, and set objective targets per campaign.
2. **Calendar Agent**
   Validate proposed dates against market holidays, blackout periods, and existing commitments. Detect conflicts and recommend alternatives.
3. **Segmentation Agent**
   Pre-build target audiences for all planned campaigns — create segment definitions, estimate sizes, and identify potential overlaps across the quarter.
4. **Content Agent**
   Start creative briefs for planned campaigns — draft thematic direction, key messages, and content requirements for early production cycles.
5. **MarTech Architecture Agent**
   Reserve infrastructure capacity for peak periods — validate send throughput, data extension capacity, and API rate limits for projected volume.

Brand Audit Cycle

Monthly audit of all active marketing materials for brand compliance. Reviews tone, colors, fonts, imagery, and terminology across every live campaign and content piece.

TRIGGER
Monthly scheduled audit (first week of each month)

OUTPUT
Brand compliance report with per-campaign scores, violation details, and corrective action plan

AGENTS INVOLVED
3 agents: Brand Guardian Agent, Campaign Manager Agent, Content Agent

STEP-BY-STEP BREAKDOWN

1. **Brand Guardian Agent**
   Audit all active marketing pieces — review tone consistency, color usage, font compliance, imagery standards, and terminology correctness.
2. **Brand Guardian Agent**
   Generate comprehensive compliance report with percentage score per campaign, detailed violation breakdowns, and severity classification.
3. **Campaign Manager Agent**
   Review violations with stakeholders, decide escalation level for each issue, and assign corrective actions to responsible agents.
4. **Content Agent**
   Correct all flagged content pieces — rewrite off-brand copy, update terminology, adjust messaging — and resubmit for review.
5. **Brand Guardian Agent**
   Re-review corrected pieces against original violations, verify fixes are adequate, and close audit with final compliance scores.

GDPR Consent Refresh

Proactive re-consent collection workflow that identifies consents approaching expiration and deploys automated re-consent journeys to maintain database compliance before regulatory deadlines.

TRIGGER
Legal agent detects consents within 30 days of expiration

OUTPUT
Re-consent journey deployed with measurement of opt-in rates and compliance status report

AGENTS INVOLVED
5 agents: Legal Agent, Segmentation Agent, Content Agent, Automation Architect Agent, Analytics Agent

STEP-BY-STEP BREAKDOWN

1. **Legal Agent**
   Detect consents approaching expiration by regulation type and market. Calculate affected contact volumes and prioritize by risk.
2. **Segmentation Agent**
   Build affected contacts segment with precise targeting — filter by consent type, market, and expiration window.
3. **Content Agent**
   Generate re-consent request messaging — clear value proposition for maintaining subscription, easy opt-in/opt-out mechanisms, regulatory-compliant language.
4. **Automation Architect Agent**
   Deploy multi-step re-consent journey — initial request, 7-day reminder, final notice with opt-out confirmation — with tracking for compliance reporting.
5. **Analytics Agent**
   Measure re-consent opt-in rate by market and segment. Report compliance status and identify at-risk populations for manual outreach.

Email Deliverability Health Check

Infrastructure and deliverability audit that reviews spam scores, authentication records (DKIM, SPF, DMARC), bounce trends, and complaint rates to maintain inbox placement.

TRIGGER
Weekly scheduled check or triggered by bounce rate exceeding 2% threshold

OUTPUT
Deliverability health report with corrective action recommendations

AGENTS INVOLVED
4 agents: QA Agent, MarTech Architecture Agent, Analytics Agent, Campaign Manager Agent

STEP-BY-STEP BREAKDOWN

1. **QA Agent**
   Run comprehensive spam score checks across all active sends. Validate DKIM signatures, SPF records, and DMARC policies for all sending domains.
2. **MarTech Architecture Agent**
   Validate infrastructure health — API response times, send queue depth, data extension query performance, and IP reputation scores.
3. **Analytics Agent**
   Report bounce rate trends (hard/soft), complaint rates by ISP, unsubscribe trends, and inbox placement rates across major providers.
4. **Campaign Manager Agent**
   Review deliverability findings, decide corrective actions (IP warmup, list cleanup, authentication fixes), and assign to responsible agents.

A/B Test Pipeline

End-to-end A/B testing workflow covering variant creation, split configuration, statistical measurement, and winner deployment. Ensures tests are properly designed with clear hypotheses and adequate sample sizes.

TRIGGER
Test hypothesis defined by Campaign Manager or Content Agent

OUTPUT
Statistically validated winner variant deployed to full audience

AGENTS INVOLVED
6 agents: Content Agent, Email Developer Agent, QA Agent, Automation Architect Agent, Analytics Agent, Campaign Manager Agent

STEP-BY-STEP BREAKDOWN

1. **Content Agent**
   Generate copy variants based on defined hypothesis — e.g., urgency vs. benefit messaging, short vs. long subject lines, personalized vs. generic CTA.
2. **Email Developer Agent**
   Build both template versions with identical layout, varying only the test element to ensure clean measurement.
3. **QA Agent**
   Validate renders for all variants across target email clients. Ensure both versions are technically identical except for the test variable.
4. **Automation Architect Agent**
   Configure split test in Journey Builder — define split percentage, test duration, winning metric, and automatic winner deployment rules.
5. **Analytics Agent**
   Monitor test results in real-time. Calculate statistical significance (95% confidence), measure lift, and produce final test report.
6. **Campaign Manager Agent**
   Review test results with stakeholders. Decide whether to deploy winner, extend test, or run follow-up test. Document learnings.

Weekly Performance Digest

Automated weekly reporting workflow that compiles KPIs across all active campaigns and channels, enriched with strategic commentary and forward-looking recommendations.

TRIGGER
Scheduled every Monday morning (automated)

OUTPUT
Executive-ready weekly performance report distributed to stakeholders

AGENTS INVOLVED
2 agents: Analytics Agent, Campaign Manager Agent

STEP-BY-STEP BREAKDOWN

1. **Analytics Agent**
   Compile weekly KPIs across all channels and campaigns — aggregate open rates, CTR, conversions, revenue, and benchmark comparisons. Identify week-over-week trends and anomalies.

2. **Campaign Manager Agent**
   Add strategic commentary — explain performance drivers, highlight wins and concerns, provide recommendations for the upcoming week, and flag campaigns requiring attention.

Template Library Refresh

Periodic audit and maintenance of the email template library. Reviews existing templates against current brand guidelines and coding standards, re-tests renders, and deprecates broken or outdated templates.

TRIGGER
Quarterly scheduled refresh or triggered by brand guideline update

OUTPUT
Updated template library with deprecated templates removed and new standards applied

AGENTS INVOLVED
3 agents: Email Developer Agent, Brand Guardian Agent, QA Agent

STEP-BY-STEP BREAKDOWN

1. **Email Developer Agent**
   Audit existing templates against current HTML/CSS standards, responsive design requirements, and dark mode compatibility. Flag outdated patterns.
2. **Brand Guardian Agent**
   Validate templates against latest brand guidelines — check colors, fonts, spacing, imagery standards, and tone alignment.
3. **QA Agent**
   Re-test all template renders across email clients. Identify broken templates and deprecate those that cannot be cost-effectively updated.

Audience Hygiene Cleanup

Data quality maintenance workflow that audits segment overlaps, validates contact data, verifies consent status, cleans stale records, and reports impact on deliverability metrics.

TRIGGER
Monthly scheduled cleanup or triggered by segment overlap exceeding 30%

OUTPUT
Clean audience data with validated consent, reduced overlaps, and improved deliverability metrics

AGENTS INVOLVED
5 agents: Segmentation Agent, CRM Intelligence Agent, Legal Agent, MarTech Architecture Agent, Analytics Agent

STEP-BY-STEP BREAKDOWN

1. **Segmentation Agent**
   Audit all active segments for overlaps exceeding 30%. Recommend deduplication strategies and segment consolidation where appropriate.
2. **CRM Intelligence Agent**
   Validate contact data quality — check for missing fields, invalid email formats, duplicate records, and stale profiles (no activity in 12+ months).
3. **Legal Agent**
   Verify consent status for all active segments — check expiration dates, consent scope, and purpose limitation compliance across markets.
4. **MarTech Architecture Agent**
   Clean data extensions — remove orphaned records, archive stale contacts, optimize table indexes, and validate referential integrity.
5. **Analytics Agent**
   Report the impact of cleanup on key metrics — deliverability improvements, bounce rate changes, audience size adjustments, and projected engagement uplift.

Platform Capabilities

Built-in features that power the agent ecosystem

Centralized Dashboard

Real-time visibility into all agents, campaigns, and departments. Workspace overview with KPI cards, status badges, and activity feeds.

Daily Operations

Automated daily standups by department, AI-generated summaries, and end-of-day reports with completed tasks, blockers, and next-day planning.

Weekly Planning Board

5-tab planning system: brainstorming, prioritization, capacity planning, resource allocation, and weekly reports with KPI tracking.

Campaign Management Hub

45+ campaign templates across lifecycle stages — acquisition, loyalty, recovery, onboarding, and communications.

AI Chat Interface

Natural language conversation with the Campaign Intelligence Coordinator via streaming chat with voice controls.

Audit & Compliance

Complete audit trail for every action: campaign launches, compliance reviews, segment creation, and legal reviews.

Unified Inbox

Centralized message inbox from all agents with filtering, task creation, and direct PM Agent communication.

Intelligence Hub

AI-generated insights, trend analysis, coverage alerts, and PM performance reports with anomaly detection.

MERKLE

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